

Miguel Camacho Horvitz

Applied economist and computational researcher

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Research profile

Applied economist and computational researcher with graduate training in economics, mathematics, and computer science. My proposed doctoral research studies how infrastructure-finance arrangements distribute fiscal and foreign-exchange risk and decision-making authority in lower-income economies, beginning with Malawi's energy sector. Previous work at the Ellison Institute of Technology, OECD, NBER, FAIR at the Norwegian School of Economics, and the World Inequality Lab spans quantitative modelling, policy evaluation, experiments, and large-scale data construction.

Education

Paris School of Economics

2021–2023

Master's Degree in Economics, Public Policies and Development

Paris, France

Selected coursework: Econometrics, advanced treatment models, political economy, development policy, public economics, labour-market policies, inequality, and machine learning in economics.

Stanford University

2015–2017

Master of Science in Computer Science

Stanford, USA

Selected coursework: Machine learning, natural-language processing, computer vision, social and information networks, algorithms, and computer systems.

Stanford University

2012–2016

Bachelor of Science in Mathematical and Computational Science

Stanford, USA

Selected coursework: Real and complex analysis, differential equations, probability, statistical inference, stochastic modelling, and convex optimisation.

Research experience

Ellison Institute of Technology, Oxford

Sep 2025–Aug 2026

Research Analyst under Prof. Andrew Scott

Oxford, United Kingdom

- Built calibrated continuous-time and life-cycle models linking health investment to survival, labour supply, and long-run growth; used numerical simulation, comparative statics, and sensitivity analysis to compare mortality reductions across age groups.
- Built a reproducible Python pipeline for UKHLS panel data and applied unsupervised learning to recover latent life-cycle health trajectories linked to income, disability, mobility, and attrition; tested robustness and translated results into policy-facing analysis.

FAIR, Norwegian School of Economics (NHH)

Sep 2024–Jun 2025

Pre-Doctoral Research Fellow under Prof. Ingar Haaland

Bergen, Norway

- Co-designed randomised field and survey experiments on demand for politically slanted news, combining treatment randomisation, generative AI, and computational text analysis to estimate effects on affective and political polarisation.
- Built an LLM-driven treatment platform and developed an open-source, Qualtrics-integrated **AI interviewer** for structured long-form qualitative data collection.

National Bureau of Economic Research

Jan–Aug 2024

Research Assistant under Prof. Ellora Derenoncourt

Cambridge, MA / Princeton, NJ

- Contributed to Martínez and Derenoncourt's paper *Market Forces and Employer Racial Preferences*, identifying data sources and study variables and supporting difference-in-differences and event-study analysis.
- Built **Python and SLURM workflows** to process approximately 34 million OCR-derived historical job advertisements, extract offered wages and candidate employer locations, geocode addresses, and run reproducible batched jobs.

OECD

Aug 2023–Sep 2025

Research Consultant

Paris / Ljubljana

- Evaluated active labour-market programmes with propensity-score matching on linked employment, health, and social-protection records; used machine learning to construct and validate a health index and estimate heterogeneous effects.

World Inequality Lab
Data Coordinator

Apr–Jul 2023
Paris, France

- Containerised workflows, migrated processing to cloud infrastructure, and integrated SQL-based preprocessing, improving the reproducibility and scalability of inequality indicators published through [WID.world](#).

ConsiliumBots

Research Fellow under Prof. Christopher Neilson

May–Sep 2022

Santiago / New York

- Supported [higher-education information interventions](#) conducted with J-PAL LAC and built machine-learning career-recommendation tools with country-specific education and labour-market data; later published an [independent synthetic-data reference implementation](#).

Selected research and policy writing

OECD report. *Does Slovenia's Public Works Programme Improve Health and Social Outcomes?*, prepared with Matija Vodopivec.

December 2025

Co-authored methods paper. *Robust Inference in the Presence of Weak Instruments*, with Kartika Herscheid and Unai Oyón-Lerga.

April 2022

Master's thesis. *On Contemporary Inequality Narratives in Chile: A comparative analysis of income and wealth distributions with respect to Chile's Household Finances Survey.*

September 2023

Co-authored policy note. *Addressing the Enduring Gender Gap in the Chilean Labour Market*, with Kartika Herscheid and Laure Lépine.

March 2022

Selected public software and replications

LLM Experimental Economics Simulation — code-first framework for reconstructing experimental-economics designs from public materials and auditing heuristic or LLM simulations against empirical targets; it does not present simulation results as empirical findings.

Colombia education CCT replication — reproducible R baseline replication of treatment effects from the Colombian conditional cash-transfer experiment using the authors' official replication package.

Basel transit investment analysis — reproducible spatial-analysis portfolio project using Swiss GTFS and Basel-Stadt open data. The grant scenario is fictional; it demonstrates data integration, geocoding, spatial joins, transit-access metrics, and validation, not a policy evaluation or recommendation.

Industry experience

Synapse

Machine Learning Engineer

2017–2020

San Francisco, USA

- Built and deployed computer-vision systems for identity verification and transaction classification in Python.
- Developed production APIs for financial risk, compliance, and fraud detection and trained generative models for adversarial testing.

Teaching and volunteering

TakenoLAB

Teacher

2025

Dzaleka Refugee Camp, Malawi

Taught introductory computer science and programming; mentored projects in web development and machine learning.

Inspirit AI

Teacher

2021–2024

Global

Taught programming and core AI concepts; mentored AI-for-social-impact projects.

Stanford University

Mathematics Tutor

2014–2016

Stanford, USA

Provided peer tutoring in linear algebra, calculus, and differential equations.

Methods, technical skills, and languages

Methods: Applied econometrics; causal inference; experimental design; panel and administrative data; structural and numerical modelling; computational text analysis; OCR/NLP; machine learning.

Programming: Python, R, Stata, SQL, Bash/Unix, C/C++, JavaScript.

Tools: AWS, GCS, OCI, Docker, Terraform, GitHub, L^AT_EX, Qualtrics.

Languages: English and Spanish (fluent); French (professional reading and writing, proficient speaking).